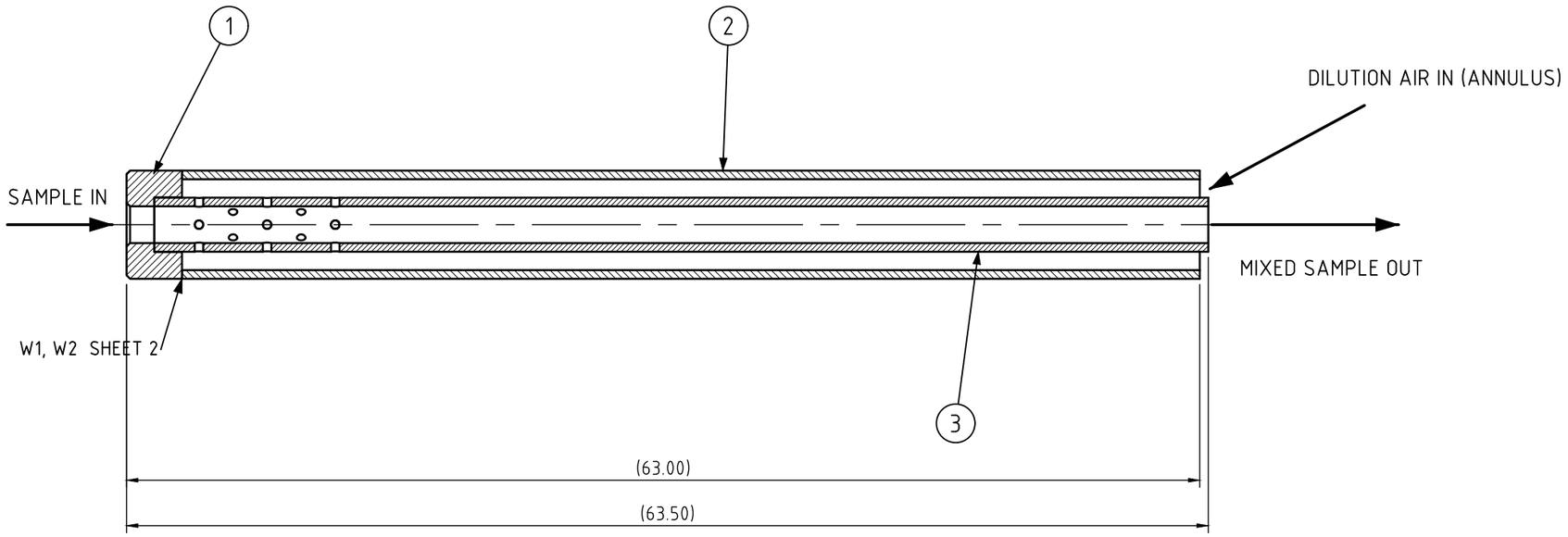
SECTION A-A TWO-TUBE PROBE ASSEMBLY

SCALE 3 : 1 ITEMS 1, 2, 3 WELDED



LIST OF MATERIAL

ITEM	QTY	PART NO.	DESCRIPTION	MATERIAL	SOURCE	SHEET
1	1	PS-RB1-SC-003	COLLAR, SEATED	316L SS BAR	UBC DESIGN	3
2	1	PS-RB1-SC-004	OUTER TUBE 1/4 X .020 WALL	316/316L SS	UBC DESIGN	4
3	1	PS-RB1-SC-005	INNER TUBE 1/8 X .020, PERFORATED	316/316L SS	UBC DESIGN	5

1. ITEMS 1-3: UBC DESIGN.
2. MATERIAL 316/316L SS (TENTATIVE); INCONEL 600/625 UNDER CONSIDERATION.
3. WETTED PARTS STAINLESS: ICAO ANNEX 16 VOL II APP 3, SAE ARP1256.
4. REAR END CONNECTIONS (DILUTION TEE, SAMPLE SEAL): PS-RB1-SCT FULL PROBE ASSEMBLY.
5. ( ) = REFERENCE.

JOINTS

MARK	JOINT	TYPE	SHEET
W1	ITEM 2 TO ITEM 1	WELD, SQUARE-GROOVE, ALL ROUND	2
W2	ITEM 3 TO ITEM 1 SOCKET	WELD, FILLET 0.20, ALL ROUND	2

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

( ) = REFERENCE SURFACE ISO 21920-1 EDGES ISO 13715 3RD ANGLE

Dept.	Technical reference	Created by	Approved by
MECH	FUSION B.1-SC v6	Siddharth Sharma	

SCALE: AS NOTED
THIRD-ANGLE PROJECTION

THE UNIVERSITY OF BRITISH COLUMBIA
Mechanical Engineering
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Document type	Document status
MFG DRAWING	PRELIMINARY

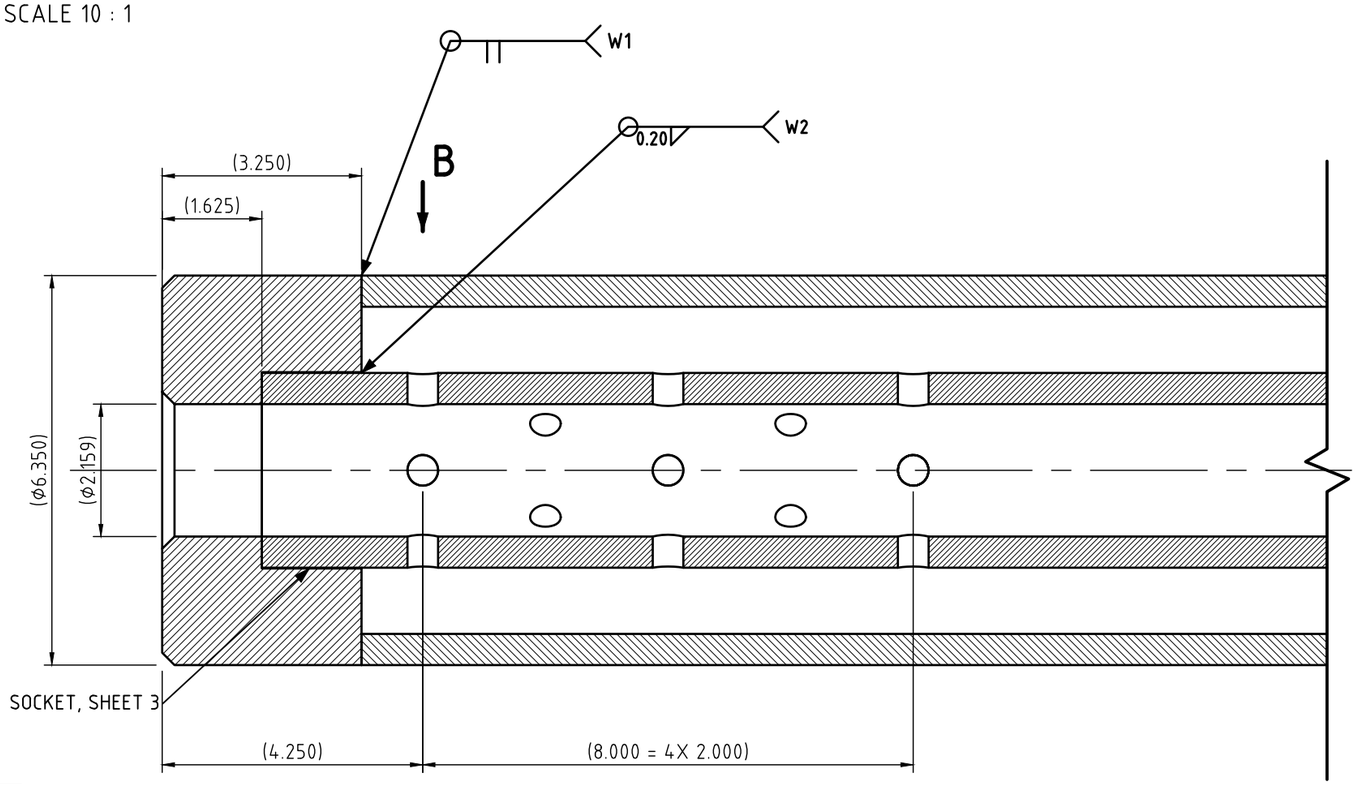
Title: GAS SAMPLING PROBE B.1-SC ASSEMBLY SECTION + LIST OF MAT.

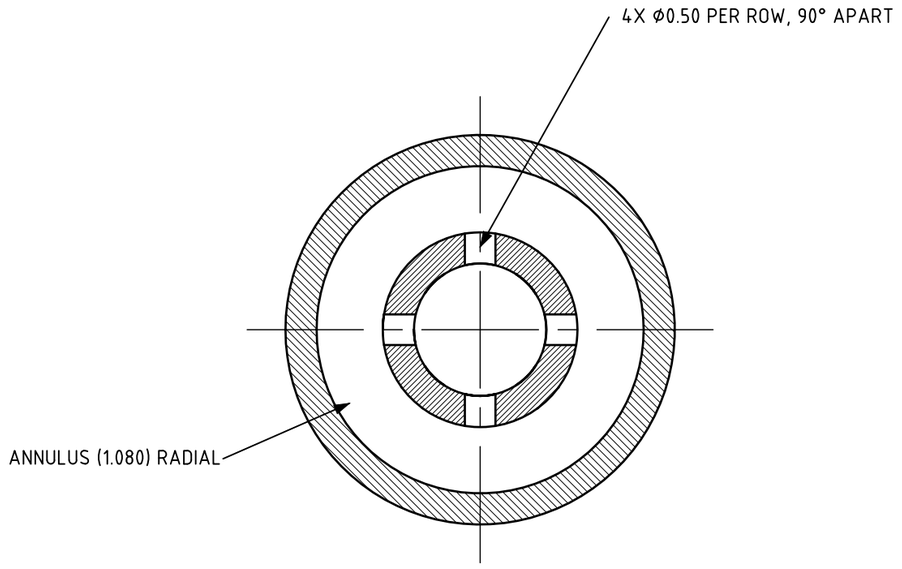
DWG No.: PS-RB1-SC-001

Rev.	Date of issue	Sheet
A	2026-10-07	1/6

DETAIL T PROBE TIP (SECTION A-A)

SCALE 10 : 1





SECTION B-B HOLE ROW 1

SCALE 10 : 1 SEEN FROM THE TIP

WELDS

MARK	JOINT	TYPE	SIZE	PROCESS / QUALITY
W2	ITEM 3 TO ITEM 1 SOCKET	FILLET, ALL ROUND	0.20 LEG MAX	MICRO-TIG / LASER, AUTOGENOUS
W1	ITEM 2 TO ITEM 1 REAR FACE	SQUARE-GROOVE, ALL ROUND	FULL PEN.	ISO 5817-B / ISO 13919-1-B

1. WELD SYMBOLS ISO 2553 SYSTEM B. ARGON PURGE, AUTOGENOUS; FILLER IF REQUIRED ER316L (AWS A5.9).
2. W2 FIRST, WITH ITEM 2 OFF. KEEP THE W2 BEAD UNDER ϕ3.600 SO ITEM 2 SLIDES ON. THEN W1.
3. NO WELD METAL IN THE BORE OR ACROSS THE ANNULUS. LEAK-CHECK W2 BEFORE W1.
4. HOLES: SHEET 5. DRILL ITEM 3 BEFORE ASSEMBLY.
5. ( ) = REFERENCE. TOLERANCES ON THE ITEM SHEETS 3 AND 5.

SEATED FIT

FEATURE	VALUE	NOTE
SOCKET, ITEM 1	ϕ3.200 +0.012/-0	REAMED, H7-LIKE
TUBE OD, ITEM 3	ϕ3.175 ±0.013	PRECISION TUBE
CLEARANCE	0.012 TO 0.050	DIAMETRAL

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

( ) = REFERENCE SURFACE ISO 21920-1 EDGES ISO 13715 3RD ANGLE

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SCALE: AS NOTED
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MFG DRAWING	PRELIMINARY

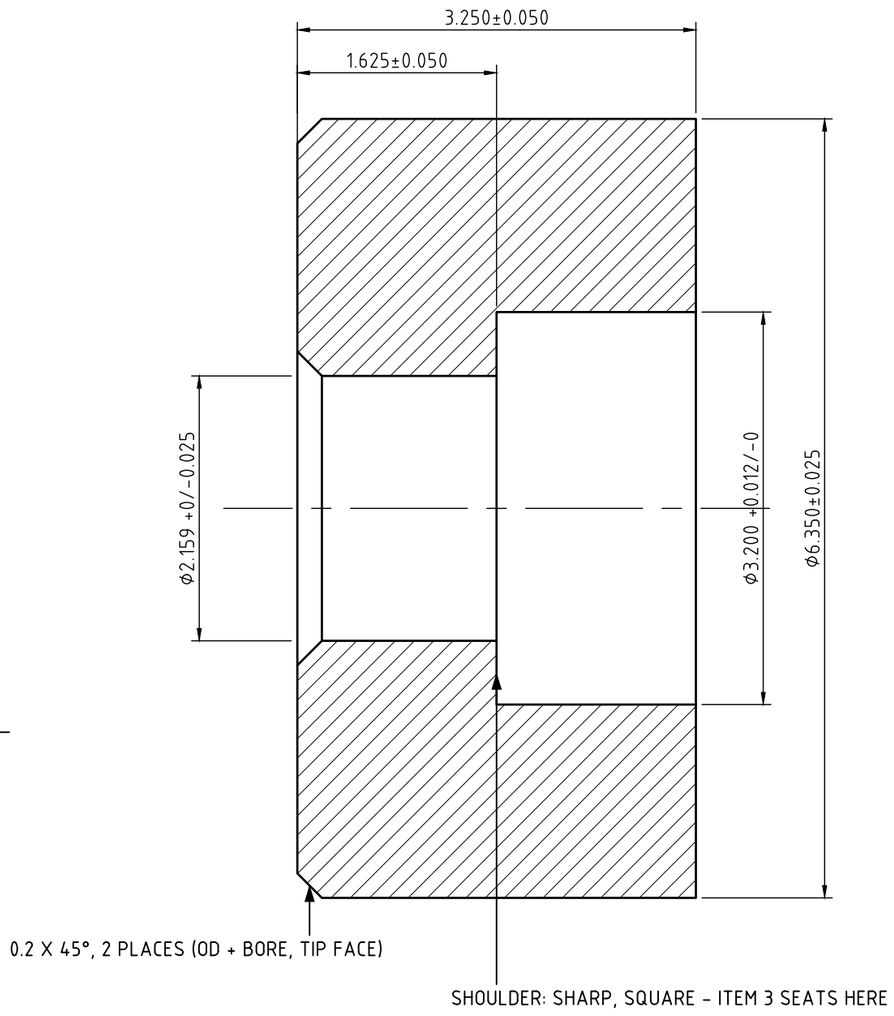
Title: GAS SAMPLING PROBE
B.1-SC PROBE TIP
SEATED JOINT + WELDS

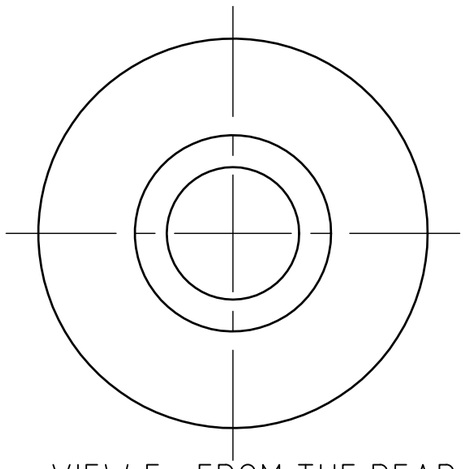
DWG No.: PS-RB1-SC-002

Rev.	Date of issue	Sheet
A	2026-10-07	2/6

SECTION A-A ITEM 1 COLLAR

SCALE 20 : 1





VIEW F FROM THE REAR

SCALE 10 : 1

COLLAR DATA

FEATURE	SIZE	TOLERANCE
COLLAR LENGTH	3.250	±0.050
TIP TO SHOULDER	1.625	±0.050
SEAT DEPTH	1.625	REF
SOCKET DIA	ϕ3.200	+0.012 / -0
BORE DIA (= ITEM 3 ID)	ϕ2.159	+0 / -0.025
COLLAR OD (= ITEM 2 OD)	ϕ6.350	±0.025
SOCKET / BORE COAXIAL	-	0.02 TIR
SHOULDER SQUARE TO BORE	-	0.02

ITEM 1

1. 316L SS BAR ϕ1/4 IN TO ASTM A479 (ALT. A276). CERTIFICATES EN 10204 3.1.
2. BORE AND SOCKET IN ONE SET-UP (0.02 TIR).
3. BREAK THE SOCKET MOUTH 0.05 MAX ONLY.
4. Ra 0.8 SOCKET + SHOULDER, Ra 1.6 BORE.
5. SOCKET FITS THE MEASURED ITEM 3 LOT: SOCKET = TUBE OD + 0.025 (SHEET 4).
6. W1, W2: SHEET 2.

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

( ) = REFERENCE SURFACE ISO 21920-1 EDGES ISO 13715 3RD ANGLE

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MFG DRAWING	PRELIMINARY

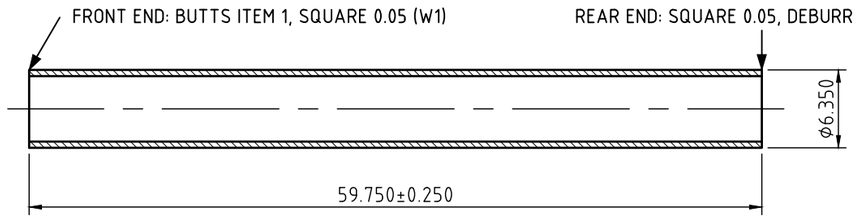
Title: GAS SAMPLING PROBE B.1-SC ITEM 1 COLLAR, SEATED

DWG No.: PS-RB1-SC-003

Rev.	Date of issue	Sheet
A	2026-10-07	3/6

ITEM 2 OUTER TUBE

SCALE 2 : 1 SECTION A-A, PLAIN TUBE



ITEM 2 – WHAT IS CONTROLLED

FEATURE	SIZE	TOLERANCE / NOTE
CUT LENGTH	59.750	±0.250
ENDS SQUARE TO AXIS	–	0.05
OD	ø6.350	AS SUPPLIED, ASTM A632
ID	(ø5.334)	REF – ANNULUS 1.080 TO ITEM 3

TUBE STOCK

ITEM	STOCK	SPEC / TEMPER	OD TOLERANCE	SUGGESTED SOURCE	ALTERNATE SOURCE
2	1/4 OD X .020 W (6.350 X 0.508)	316 SMLS, ANNEALED, A632	+0.004/–0 IN	MICROGROUP 316F20250X020SL	VITA NEEDLE; ACCU-TUBE
3	1/8 OD X .020 W (3.175 X 0.508)	316 SMLS, ANNEALED, A632	±0.013 mm (NOTE 2)	MICROGROUP 316F20125X020SL	VITA NEEDLE; ACCU-TUBE
ALT	SAME AS 2, 3	INCONEL 600 (B167) / 625 (B444)	–	MICROGROUP (ALLOY 600)	–

1. ALL TUBES SEAMLESS 316/316L, FULLY ANNEALED (90 HRB MAX). CERTIFICATES EN 10204 3.1.
2. ITEM 3: IF THE MEASURED TUBE OD EXCEEDS ø3.188, MACHINE THE ITEM 1 SOCKET TO TUBE OD + 0.025 (SHEET 3).
3. MATERIAL 316 IS TENTATIVE; INCONEL 600/625 UNDER CONSIDERATION FOR HIGHER TIP TEMPERATURE.
4. CUT, DEBURR INSIDE AND OUT, CLEAN AND DRY. NO MACHINING OF OD OR ID.

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

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MFG DRAWING	PRELIMINARY

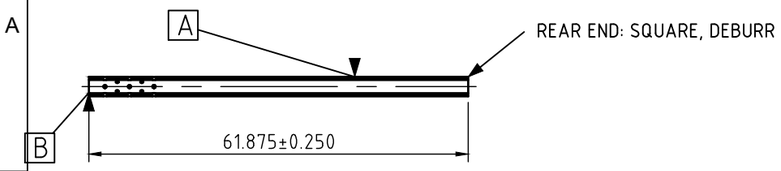
Title: GAS SAMPLING PROBE
B.1-SC ITEM 2
OUTER TUBE + STOCK

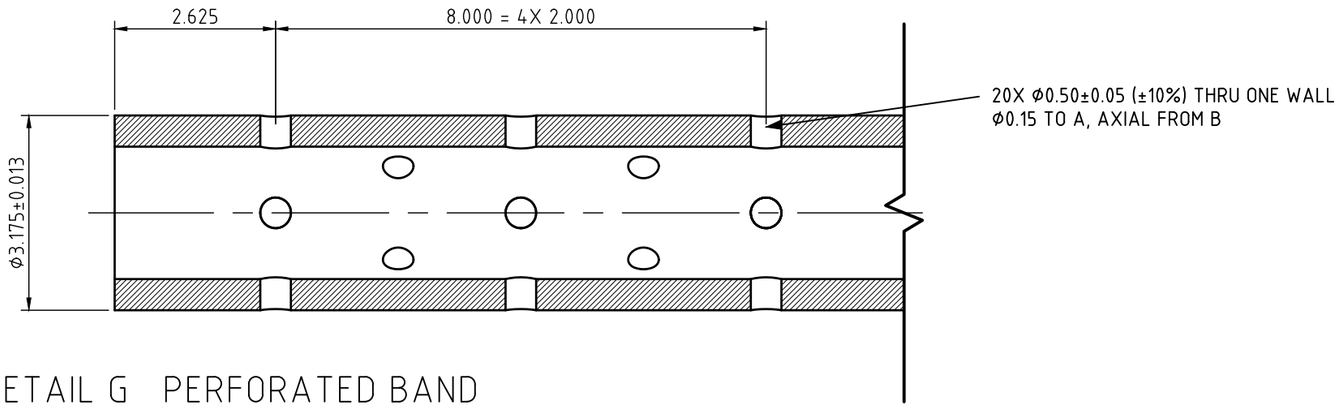
DWG No.: PS-RB1-SC-004

Rev.	Date of issue	Sheet
A	2026-10-07	4/6

ITEM 3 INNER TUBE, PERFORATED

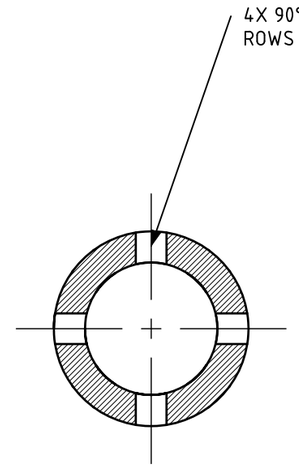
SCALE 1 : 1 SECTION A–A





DETAIL G PERFORATED BAND

SCALE 10 : 1 ROWS 1, 3, 5 IN THE CUT



SECTION B–B ROW 1

SCALE 10 : 1 ITEM 3 ONLY

HOLE PATTERN – ITEM 3

FEATURE	VALUE
TOTAL HOLES	20 = 5 ROWS X 4
HOLE DIAMETER	Ø0.50 ±0.05 (±10 %)
AXIAL PITCH	2.000
ROW 1 FROM SEATED END B	2.625 (4.250 FROM TIP)
ANGULAR PITCH / ROW CLOCKING	90° / 45°
SUM OF HOLE AREAS, NOMINAL	3.93 mm2
SAMPLE INLET AREA, Ø2.159	3.66 mm2
EQUAL-AREA HOLE DIAMETER	(Ø0.483)

DESIGN INTENT
SUM OF ALL HOLE AREAS = SAMPLE INLET AREA
(Ø2.159 TIP BORE, 3.66 mm2).
NOMINAL Ø0.50: 107 %. LIMITS Ø0.45–0.55: 87–130 %.
MEASURE ALL 20 HOLES AND REPORT THE SUM.

1. LASER DRILL, TUBE FREE, BEFORE ASSEMBLY (SAE AMS2548 OR EQUIV.).
2. CYLINDRICAL: ENTRY AND EXIT DIA WITHIN 0.05 – NO TAPER.
3. ONE WALL ONLY. DEBURR, NO RECAST OR BURR IN THE BORE.
4. 100 % COUNT; EACH HOLE GO Ø0.45 / NO-GO Ø0.55.
5. TUBE STOCK: SHEET 4.

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

( ) = REFERENCE SURFACE ISO 21920-1 EDGES ISO 13715 3RD ANGLE

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Title: GAS SAMPLING PROBE
B.1-SC ITEM 3
INNER TUBE + HOLES

DWG No.: PS-RB1-SC-005

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A	2026-10-07	5/6

ASSEMBLY SEQUENCE

STEP	OPERATION	SHEET
1	LASER DRILL ITEM 3 (TUBE FREE); INSPECT ALL HOLES	5
2	SEAT ITEM 3 IN ITEM 1 ON A MANDREL; WELD W2	2
3	CHECK W2 BEAD + RUNOUT; SLIDE ITEM 2 ON; WELD W1	2
4	PROOF + LEAK TEST W1 AND W2	6
5	CLEAN, DRY, CAP BOTH ENDS FOR SHIPPING	-

INSPECTION

NO	ITEM	CHARACTERISTIC	REQUIREMENT	METHOD
1	1	SOCKET DIA	⌀3.200 +0.012/−0 (OR LOT OD+0.025)	PLUG GAUGE
2	1	TIP TO SHOULDER	1.625 ±0.050	DEPTH GAUGE
3	1	BORE DIA	⌀2.159 +0/−0.025	PIN GAUGE
4	1	SOCKET / BORE COAXIAL	0.02 TIR	INDICATOR
5	3	TUBE OD	⌀3.175 ±0.013	MICROMETER
6	3	HOLE DIA, EACH	⌀0.50 ±0.05	GO / NO-GO
7	3	HOLE COUNT	20 = 5 X 4	COUNT
8	3	SUM OF HOLE AREAS	REPORT; TARGET 3.66 mm2	OPTICAL
9	3	NO TAPER	ENTRY − EXIT 0.05 MAX	OPTICAL, BOTH SIDES
10	3	CUT LENGTH	61.875 ±0.250	CALIPER
11	2	CUT LENGTH	59.750 ±0.250	CALIPER
12	ASSY	W1, W2	ISO 5817-B / 13919-1-B, LEAK-TIGHT	VISUAL 10X + LEAK
13	ASSY	ITEM 3 RUNOUT	⌀0.20 TO ITEM 1 OD	INDICATOR
14	ASSY	REAR ENDS FROM TIP	63.00 / 63.50 ±0.25	CALIPER

TESTS

TEST	REQUIREMENT	REFERENCE
PROOF, PNEUMATIC	1.1 X DESIGN PRESSURE, 10 MIN HOLD	ASME B31.3 345.5 PRACTICE
LEAK, W1 + W2	NO BUBBLES AT PROOF PRESSURE	ASTM E515
LEAK, OPTIONAL	HELIUM, TIP WELDS	ASTM E499

STANDARDS

AREA	STANDARD
SAMPLING	SAE ARP1256E; ICAO ANNEX 16 VOL II APP 3; 40 CFR 1065 (PRACTICE)
TUBE	ASTM A632 (ALT. A269); INCONEL: ASTM B167 (600), B444 (625)
BAR	ASTM A479 (ALT. A276)
CERTIFICATES	EN 10204 3.1
WELDING	ISO 2553 SYSTEM B SYMBOLS; ISO 5817-B; ISO 13919-1-B; AWS A5.9 ER316L
HOLES	SAE AMS2548 LASER BEAM MACHINING, OR EQUIVALENT
PRESSURE, LEAK	ASME B31.3 345.5 (PRACTICE); ASTM E515; ASTM E499
DRAWING	ISO 128, 129-1, 2768-m, 22081, 8015, 1101, 21920-1, 13715

1. CLEAN ALL WETTED PARTS FOR EMISSIONS SAMPLING: DRY, OIL-FREE.
2. RECORD TUBE LOT OD, SOCKET DIA AND HOLE SUM IN THE BUILD RECORD.
3. DESIGN PRESSURE: AS SPECIFIED BY UBC.

GENERAL TOL. ISO 2768-m GEOMETRY ISO 22081 TOLERANCING ISO 8015 mm

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SCALE: AS NOTED THIRD-ANGLE PROJECTION		Document type MFG DRAWING	Document status PRELIMINARY
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			Rev. A / Date of issue 2026-10-07 / Sheet 6/6